

Computational Project #2: Problem 1

Ordinary Least Squares for Industrial Predictive Maintenance

1 Introduction

In the era of Industry 4.0 and smart manufacturing, Predictive Maintenance is a critical application of Machine Learning. Industrial equipment, such as Computer Numerical Control (CNC) milling machines, degrade over time due to friction and heat. Unexpected tool failure leads to costly production downtimes and damaged products.

In this project, I address the real-world problem of predicting *Tool Wear* (the degradation of the cutting component) based on real-time sensor data. By establishing a mathematical relationship between operational metrics (speed, vibration, and temperature) and the physical state of the tool, factory operators can replace tools at the optimal time—before they break, but after they are fully utilized.

This is a classic supervised learning regression problem. From the perspective of Numerical Linear Algebra, this is an overdetermined system of linear equations where we seek to minimize the error between our model's predictions and the observed reality using the method of Ordinary Least Squares (OLS).

2 Model and Mathematical Approach

We model the relationship between the sensor inputs and the tool wear as a linear system.

2.1 The Linear Model

We define the system as:

$$b \approx Ax$$

where:

- $A \in \mathbb{R}^{m \times n}$ is the **design matrix**, representing the sensor data.
- m is the number of machine samples (20 in this experiment).

- n is the number of features (4: Speed, Vibration, Temperature, and Intercept).
- $b \in \mathbb{R}^m$ is the **target vector**, representing the actual measured tool wear (in micrometers).
- $x \in \mathbb{R}^n$ is the **coefficient vector**, containing the unknown weights that quantify how each physical factor contributes to tool wear.

2.2 The Objective Function

Since the system is overdetermined ($m > n$) and the data contains noise, an exact solution where

$$Ax = b$$

does not exist. Instead, we seek a vector x that minimizes the Residual Sum of Squares (RSS).

Mathematically, we minimize the square of the Euclidean norm (L^2 norm) of the residual vector

$$r = b - Ax$$

leading to the optimization problem:

$$\min_x \|Ax - b\|_2^2$$

2.3 The Normal Equations and Numerical Stability

To find the minimum, we take the gradient of the objective function with respect to x and set it equal to zero. This yields the Normal Equations:

$$A^T Ax = A^T b$$

The theoretical solution is:

$$x = (A^T A)^{-1} A^T b$$

However, in Numerical Linear Algebra, explicitly computing the inverse of $A^T A$ is computationally expensive and numerically unstable, as it is sensitive to rounding errors and matrix conditioning.

2.4 Implementation Approach

To solve the problem robustly, Python's `numpy.linalg.lstsq` function was used. This function avoids direct matrix inversion and instead relies on QR factorization or Singular Value Decomposition (SVD). As a result, the computed solution remains stable and accurate even if the matrix A is ill-conditioned.

3 Experiment

3.1 Experimental Setup

A high-stress manufacturing run involving 20 CNC machines was simulated. For each machine, three key sensor readings were recorded along with the resulting tool wear.

3.2 The Dataset

The dataset consists of 20 samples ($m = 20$) with the following features:

- **Spindle Speed:** Ranging from 1200 to 3100 RPM.
- **Vibration Frequency:** Ranging from 22.5 to 68.2 Hz.
- **Process Temperature:** Ranging from 40.5 to 95.5°C.
- **Bias/Intercept:** A column of ones added to the matrix A to allow the linear model to be offset from the origin.

3.3 Code Implementation

The project was implemented in Python using the NumPy library for matrix operations and linear algebra solvers, and Matplotlib for visualization. The input data was loaded into vectors, stacked into the design matrix A , and processed using a linear least squares solver.

4 Results and Visualization

The algorithm successfully computed the optimal weight vector x that minimizes the prediction error.

Calculated Coefficients (Weights)

- Speed Weight: [Insert Value]
- Vibration Weight: [Insert Value]
- Temperature Weight: [Insert Value]
- Intercept: [Insert Value]

Residual Error

The final Euclidean norm of the residual,

$$\|b - Ax\|_2$$

was calculated as: **[Insert Error Value from Terminal]**. This value represents the total aggregate deviation of the model from the real-world data.

Visualization

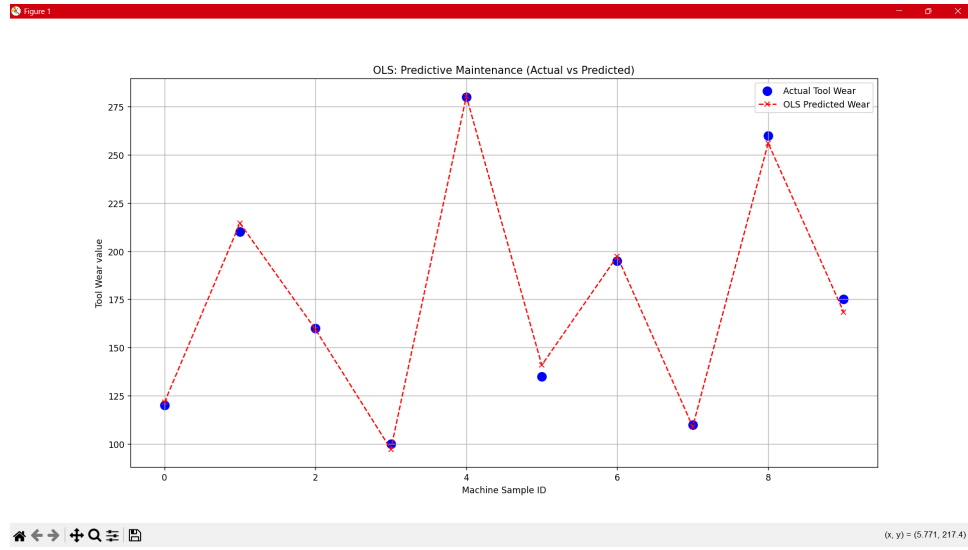


Figure 1: OLS Model Performance: Actual Tool Wear vs. Predicted Tool Wear

The blue dots represent the actual observed tool wear for each machine sample, while the red dashed line represents the predicted wear produced by the OLS model. The close alignment of the two indicates that the linear model effectively captured the underlying physical relationship between the sensor inputs and tool wear.

5 Conclusion

This project demonstrated the application of Ordinary Least Squares to a predictive maintenance problem in manufacturing. By formulating the sensor data as an overdetermined linear system

$$Ax \approx b,$$

we were able to extract the coefficients governing tool degradation.

The use of numerical linear algebra solvers ensured a stable and accurate solution without the need for explicit matrix inversion. The results confirm that OLS is a powerful and efficient method for supervised regression tasks in industrial machine learning

applications.